

Matrices and Systems of Linear Equations (2015–2021)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS

32

TOTAL MARKS

59 M

TEST DURATION

96 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	14	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	6	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	12	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q59 • ID: JAM_2021_Q59)

NAT

+2 M

Neg: Nil (0)

Q. 59 Let $A = \begin{pmatrix} 2 & -1 & 3 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{pmatrix}$. Then the largest eigenvalue of A is ____.

Question 2 (JAM 2021 • Q56 • ID: JAM_2021_Q56)

NAT

+2 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 56 The determinant of the matrix

$$\begin{pmatrix} 2021 & 2020 & 2020 & 2020 \\ 2021 & 2021 & 2020 & 2020 \\ 2021 & 2021 & 2021 & 2020 \\ 2021 & 2021 & 2021 & 2021 \end{pmatrix}$$

is ____.

Question 3 (JAM 2021 • Q40 • ID: JAM_2021_Q40)

MSQ

+2 M

Neg: Nil (0)

Q. 40 Let $m > 1$ and $n > 1$ be integers. Let A be an $m \times n$ matrix such that for some $m \times 1$ matrix b_1 , the equation $Ax = b_1$ has infinitely many solutions. Let b_2 denote an $m \times 1$ matrix different from b_1 . Then $Ax = b_2$ has

- (A) infinitely many solutions for some b_2 . (B) a unique solution for some b_2 .
 (C) no solution for some b_2 . (D) finitely many solutions for some b_2 .



MA

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Question 4 (JAM 2021 • Q22 • ID: JAM_2021_Q22)

MCQ

+2 M

Neg: -0.67

Q. 22 Let A be an $n \times n$ invertible matrix and C be an $n \times n$ nilpotent matrix. If $X = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix}$ is a $2n \times 2n$ matrix (each X_{ij} being $n \times n$) that commutes with the $2n \times 2n$ matrix $B = \begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$, then

- (A) X_{11} and X_{22} are necessarily zero matrices.
- (B) X_{12} and X_{21} are necessarily zero matrices.
- (C) X_{11} and X_{21} are necessarily zero matrices.
- (D) X_{12} and X_{22} are necessarily zero matrices.

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Question 5 (JAM 2021 • Q10 • ID: JAM_2021_Q10)

MCQ

+1 M

Neg: -0.33

JAM 2021

MATHEMATICS - MA

Q. 10 Let $n > 1$ be an integer. Consider the following two statements for an arbitrary $n \times n$ matrix A with complex entries.

- I. If $A^k = I_n$ for some integer $k \geq 1$, then all the eigenvalues of A are k^{th} roots of unity.
II. If, for some integer $k \geq 1$, all the eigenvalues of A are k^{th} roots of unity, then $A^k = I_n$.

Then

- (A) both I and II are TRUE. (B) I is TRUE but II is FALSE.
(C) I is FALSE but II is TRUE. (D) neither I nor II is TRUE.



MA

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Question 6 (JAM 2020 • Q58 • ID: JAM_2020_Q58)

NAT

+2 M

Neg: Nil (0)

Q. 58 Let $M = \begin{bmatrix} 9 & 2 & 7 & 1 \\ 0 & 7 & 2 & 1 \\ 0 & 0 & 11 & 6 \\ 0 & 0 & -5 & 0 \end{bmatrix}$. Then, the value of $\det((8I - M)^3)$ is _____.

Question 7 (JAM 2020 • Q36 • ID: JAM_2020_Q36)

MSQ

+2 M

Neg: Nil (0)

Q. 36 Consider the following system of linear equations

$$x + y + 5z = 3, \quad x + 2y + mz = 5 \quad \text{and} \quad x + 2y + 4z = k.$$

The system is consistent if

(A) $m \neq 4$ (B) $k \neq 5$ (C) $m = 4$ (D) $k = 5$

MA

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Question 8 (JAM 2020 • Q23 • ID: JAM_2020_Q23)

MCQ

+2 M

Neg: -0.67

Q. 23 Let M be a 4×3 real matrix and let $\{e_1, e_2, e_3\}$ be the standard basis of $\mathbb{R}^3$. Which of the following is true?

(A) If $\text{rank}(M) = 1$, then $\{Me_1, Me_2\}$ is a linearly independent set(B) If $\text{rank}(M) = 2$, then $\{Me_1, Me_2\}$ is a linearly independent set(C) If $\text{rank}(M) = 2$, then $\{Me_1, Me_3\}$ is a linearly independent set(D) If $\text{rank}(M) = 3$, then $\{Me_1, Me_3\}$ is a linearly independent set

Question 9 (JAM 2020 • Q20 • ID: JAM_2020_Q20)

MCQ

+2 M

Neg: -0.67

Q. 20 Let M be an $n \times n$ ($n \geq 2$) non-zero real matrix with $M^2 = 0$ and let $\alpha \in \mathbb{R} \setminus \{0\}$. Then

- (A) α is the only eigenvalue of $(M + \alpha I)$ and $(M - \alpha I)$
 (B) α is the only eigenvalue of $(M + \alpha I)$ and $(\alpha I - M)$
 (C) $-\alpha$ is the only eigenvalue of $(M + \alpha I)$ and $(M - \alpha I)$
 (D) $-\alpha$ is the only eigenvalue of $(M + \alpha I)$ and $(\alpha I - M)$

MA

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Question 10 (JAM 2020 • Q19 • ID: JAM_2020_Q19)

MCQ

+2 M

Neg: -0.67

Q. 19 Let M be a real 6×6 matrix. Let 2 and -1 be two eigenvalues of M . If $M^5 = aI + bM$, where $a, b \in \mathbb{R}$, then

- (A) $a = 10, b = 11$ (B) $a = -11, b = 10$
 (C) $a = -10, b = 11$ (D) $a = 10, b = -11$

Question 11 (JAM 2019 • Q58 • ID: JAM_2019_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 Let M be a 3×3 matrix with real entries such that $M^2 = M + 2I$, where I denotes the 3×3 identity matrix. If α, β and γ are eigenvalues of M such that $\alpha\beta\gamma = -4$, then $\alpha + \beta + \gamma$ is equal to _____

Question 12 (JAM 2019 • Q57 • ID: JAM_2019_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 Let M and N be any two 4×4 matrices with integer entries satisfying

$$MN = 2 \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then the maximum value of $\det(M) + \det(N)$ is _____

Question 13 (JAM 2019 • Q56 • ID: JAM_2019_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 If $\begin{pmatrix} 2 \\ y \\ z \end{pmatrix}$, $y, z \in \mathbb{R}$, is an eigenvector corresponding to a real eigenvalue of the matrix $\begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -4 \\ 0 & 1 & 3 \end{pmatrix}$ then $z - y$ is equal to _____

Question 14 (JAM 2019 • Q43 • ID: JAM_2019_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 Consider the following system of three linear equations in four unknowns x_1, x_2, x_3 and x_4

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 4, \\ x_1 + 2x_2 + 3x_3 + 4x_4 &= 5, \\ x_1 + 3x_2 + 5x_3 + kx_4 &= 5. \end{aligned}$$

If the system has no solutions, then $k =$ _____

Question 15 (JAM 2019 • Q27 • ID: JAM_2019_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 The set of eigenvalues of which one of the following matrices is NOT equal to the set of eigenvalues of $\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$?

(A) $\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$

(B) $\begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$

Question 16 (JAM 2019 • Q12 • ID: JAM_2019_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let P be a 4×4 matrix with entries from the set of rational numbers. If $\sqrt{2} + i$, with $i = \sqrt{-1}$, is a root of the characteristic polynomial of P and I is the 4×4 identity matrix, then

(A) $P^4 = 4P^2 + 9I$ (B) $P^4 = 4P^2 - 9I$ (C) $P^4 = 2P^2 - 9I$ (D) $P^4 = 2P^2 + 9I$

Question 17 (JAM 2018 • Q37 • ID: JAM_2018_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37 Let $m, n \in \mathbb{N}$, $m < n$, $P \in M_{n \times m}(\mathbb{R})$, $Q \in M_{m \times n}(\mathbb{R})$. Then which of the following is (are) NOT possible?

- (A) $\text{rank}(PQ) = n$
- (B) $\text{rank}(QP) = m$
- (C) $\text{rank}(PQ) = m$
- (D) $\text{rank}(QP) = \left\lceil \frac{m+n}{2} \right\rceil$, the smallest integer larger than or equal to $\frac{m+n}{2}$

Question 18 (JAM 2018 • Q36 • ID: JAM_2018_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36 Suppose $\alpha, \beta, \gamma \in \mathbb{R}$. Consider the following system of linear equations.

$x + y + z = \alpha$, $x + \beta y + z = \gamma$, $x + y + \alpha z = \beta$. If this system has at least one solution, then which of the following statements is (are) TRUE?

- (A) If $\alpha = 1$ then $\gamma = 1$
- (B) If $\beta = 1$ then $\gamma = \alpha$
- (C) If $\beta \neq 1$ then $\alpha = 1$
- (D) If $\gamma = 1$ then $\alpha = 1$

Question 19 (JAM 2018 • Q27 • ID: JAM_2018_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 Let I denote the 4×4 identity matrix. If the roots of the characteristic polynomial of a 4×4 matrix

M are $\pm \sqrt{\frac{1 \pm \sqrt{5}}{2}}$, then $M^8 =$

- (A) $I + M^2$
- (B) $2I + M^2$
- (C) $2I + 3M^2$
- (D) $3I + 2M^2$

Question 20 (JAM 2017 • Q58 • ID: JAM_2017_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 Let $\alpha, \beta, \gamma, \delta$ be the eigenvalues of the matrix

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Then $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 =$ _____

Question 21 (JAM 2017 • Q48 • ID: JAM_2017_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let P be a 7×7 matrix of rank 4 with real entries. Let $\mathbf{a} \in \mathbb{R}^7$ be a column vector. Then the rank of $P + \mathbf{a}\mathbf{a}^T$ is at least _____

Question 22 (JAM 2017 • Q38 • ID: JAM_2017_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38 Let M be an $n \times n$ matrix with real entries such that $M^3 = I$. Suppose that $M\mathbf{v} \neq \mathbf{v}$ for any non-zero vector $\mathbf{v}$. Then which of the following statements is / are TRUE?

- (A) M has real eigenvalues
- (B) $M + M^{-1}$ has real eigenvalues
- (C) n is divisible by 2
- (D) n is divisible by 3

Question 23 (JAM 2017 • Q34 • ID: JAM_2017_Q34)

MSQ

+2 M

Neg: Nil (0)

Q.34 If X and Y are $n \times n$ matrices with real entries, then which of the following is/are TRUE?

- (A) If $P^{-1}XP$ is diagonal for some real invertible matrix P , then there exists a basis for $\mathbb{R}^n$ consisting of eigenvectors of X
- (B) If X is diagonal with distinct diagonal entries and $XY = YX$, then Y is also diagonal
- (C) If X^2 is diagonal, then X is diagonal
- (D) If X is diagonal and $XY = YX$ for all Y , then $X = \lambda I$ for some $\lambda \in \mathbb{R}$

Question 24 (JAM 2017 • Q12 • ID: JAM_2017_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let $\mathcal{M}$ be the set of all invertible 5×5 matrices with entries 0 and 1. For each $M \in \mathcal{M}$, let $n_1(M)$ and $n_0(M)$ denote the number of 1's and 0's in M , respectively. Then

$$\min_{M \in \mathcal{M}} |n_1(M) - n_0(M)| =$$

- (A) 1
- (B) 3
- (C) 5
- (D) 15

Question 25 (JAM 2016 • Q53 • ID: JAM_2016_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Let

$$M = \begin{bmatrix} \alpha & 1 & 1 \\ 1 & \beta & 1 \\ 1 & 1 & \gamma \end{bmatrix}, \quad \alpha\beta\gamma = 1, \quad \alpha, \beta, \gamma \in \mathbb{R} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3.$$

Then $M\mathbf{x} = \mathbf{0}$ has infinitely many solutions if $\text{trace}(M)$ is _____

Question 26 (JAM 2016 • Q43 • ID: JAM_2016_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3$ be a non-zero vector and $A = \frac{\mathbf{x} \mathbf{x}^T}{\mathbf{x}^T \mathbf{x}}$. Then the dimension of the vector space $\{\mathbf{y} \in \mathbb{R}^3 \mid A\mathbf{y} = \mathbf{0}\}$ over $\mathbb{R}$ is _____

Question 27 (JAM 2016 • Q27 • ID: JAM_2016_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 Let $M = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & \alpha \\ 2 & -\alpha & 0 \end{bmatrix}$, $\alpha \in \mathbb{R} \setminus \{0\}$ and $\mathbf{b}$ a non-zero vector such that $M\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} \in \mathbb{R}^3$. Then the value of $\mathbf{x}^T \mathbf{b}$ is

(A) $-\alpha$ (B) α (C) 0 (D) 1

Question 28 (JAM 2016 • Q26 • ID: JAM_2016_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 The matrix $M = \begin{bmatrix} \cos \alpha & \sin \alpha \\ i \sin \alpha & i \cos \alpha \end{bmatrix}$ is a unitary matrix when α is

(A) $(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$ (B) $(3n+1)\frac{\pi}{3}, n \in \mathbb{Z}$

(C) $(4n+1)\frac{\pi}{4}, n \in \mathbb{Z}$ (D) $(5n+1)\frac{\pi}{5}, n \in \mathbb{Z}$

Question 29 (JAM 2016 • Q13 • ID: JAM_2016_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 The largest eigenvalue of the matrix $\begin{bmatrix} 1 & 4 & 16 \\ 4 & 16 & 1 \\ 16 & 1 & 4 \end{bmatrix}$ is

(A) 16 (B) 21

(C) 48 (D) 64

Question 30 (JAM 2015 • Q60 • ID: JAM_2015_Q60)

NAT

+2 M

Neg: Nil (0)

Q.20 Let P and Q be two real matrices of size 4×6 and 5×4 , respectively. If $\text{rank}(Q) = 4$ and $\text{rank}(QP) = 2$, then $\text{rank}(P)$ is equal to _____

END OF THE QUESTION PAPER

MA

16/16

Question 31 (JAM 2015 • Q16 • ID: JAM_2015_Q16)

MCQ

+2 M

Neg: -0.67

Q.16

Let $A = \begin{bmatrix} 0 & 1-i \\ -1-i & i \end{bmatrix}$ and $B = A^T \bar{A}$. Then

- (A) an eigenvalue of B is purely imaginary
- (B) an eigenvalue of A is zero
- (C) all eigenvalues of B are real
- (D) A has a non-zero real eigenvalue

Question 32 (JAM 2015 • Q3 • ID: JAM_2015_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let a, b, c, d be distinct non-zero real numbers with $a + b = c + d$. Then an eigenvalue of the matrix $\begin{bmatrix} a & b & 1 \\ c & d & 1 \\ 1 & -1 & 0 \end{bmatrix}$ is

- (A) $a + c$
- (B) $a + b$
- (C) $a - b$
- (D) $b - d$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Matrices and Systems of Linear Equations (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	4
2	NAT	+2	2021
3	MSQ	+2	MTA
4	MCQ	+2	B
5	MCQ	+1	B
6	NAT	+2	−216 to −216
7	MSQ	+2	A;D
8	MCQ	+2	D
9	MCQ	+2	B
10	MCQ	+2	A
11	NAT	+2	3 to 3

Q. No.	Type	Marks	Official Key
12	NAT	+2	17 to 17
13	NAT	+2	3 to 3
14	NAT	+1	7 to 7
15	MCQ	+2	D
16	MCQ	+2	C
17	MSQ	+2	A;D
18	MSQ	+2	A;B
19	MCQ	+2	C
20	NAT	+2	5.9 to 6.1
21	NAT	+1	2.9 to 3.1
22	MSQ	+2	B, C

Q. No.	Type	Marks	Official Key
23	MSQ	+2	A, B, D
24	MCQ	+2	A
25	NAT	+2	3.0 to 3.0
26	NAT	+1	2.0 to 2.0
27	MCQ	+2	C
28	MCQ	+2	A
29	MCQ	+2	B
30	NAT	+2	2
31	MCQ	+2	C
32	MCQ	+1	B

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.